

```
!pip -q install opencv-python-headless

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
from google.colab.patches import cv2_imshow
import os

print("OpenCV version:", cv2.__version__)
```

OpenCV version: 5.0.0

```
uploaded = files.upload()

image_name = next(iter(uploaded.keys()))
image = cv2.imread(image_name)

if image is None:
    raise ValueError("The uploaded file is not a valid image.")

print("Image loaded successfully:", image_name)
print("Image shape:", image.shape)
```

cat.jpg
cat.jpg(image/jpeg) - 56967 bytes, last modified: 8/24/2026 - 100% done
Saving cat.jpg to cat.jpg
Image loaded successfully: cat.jpg
Image shape: (873, 662, 3)

```
print("Display using OpenCV:")
cv2_imshow(image)

image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(8, 6))
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Display using OpenCV:



Original Image



```
height, width, channels = image.shape
```

```
print("Image Properties")
print("Width:", width)
print("Height:", height)
print("Channels:", channels)
print("Data Type:", image.dtype)
print("Total Pixels:", width * height)
```

Image Properties
Width: 662
Height: 873
Channels: 3
Data Type: uint8
Total Pixels: 577926

```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(8, 6))
plt.imshow(gray, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")
plt.show()
```

Grayscale Image



```
hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)

plt.figure(figsize=(8, 6))
plt.imshow(cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB))
plt.title("HSV Image")
plt.axis("off")
plt.show()
```

HSV Image



```
lab = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)

lab_rgb = cv2.cvtColor(lab, cv2.COLOR_LAB2RGB)

plt.figure(figsize=(8, 6))
plt.imshow(lab_rgb)
plt.title("LAB Image")
plt.axis("off")
plt.show()
```

LAB Image



```
resized = cv2.resize(image, (400, 400))

plt.figure(figsize=(6, 6))
plt.imshow(cv2.cvtColor(resized, cv2.COLOR_BGR2RGB))
plt.title("Resized Image")
plt.axis("off")
plt.show()
```

Resized Image



```
height, width = image.shape[:2]

center = (width // 2, height // 2)
matrix = cv2.getRotationMatrix2D(center, 90, 1)

rotated = cv2.warpAffine(image, matrix, (width, height))

plt.figure(figsize=(8, 6))
plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))
plt.title("Rotated Image - 90 Degrees")
plt.axis("off")
plt.show()
```

Rotated Image - 90 Degrees



```
horizontal_flip = cv2.flip(image, 1)
vertical_flip = cv2.flip(image, 0)

plt.figure(figsize=(8, 6))
plt.imshow(cv2.cvtColor(horizontal_flip, cv2.COLOR_BGR2RGB))
plt.title("Horizontal Flip")
plt.axis("off")
```

```
plt.show()

plt.figure(figsize=(8, 6))
plt.imshow(cv2.cvtColor(vertical_flip, cv2.COLOR_BGR2RGB))
plt.title("Vertical Flip")
plt.axis("off")
plt.show()
```

Horizontal Flip



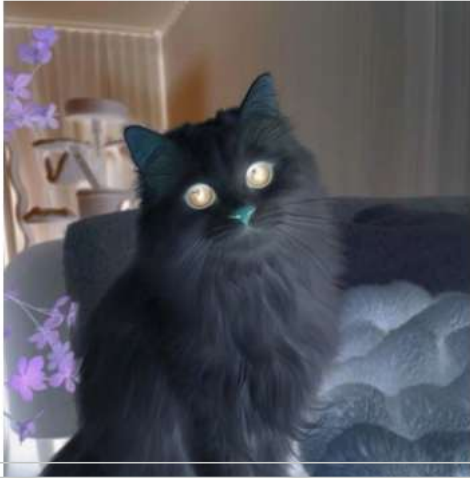
Vertical Flip



```
negative = 255 - image

plt.figure(figsize=(8, 6))
plt.imshow(cv2.cvtColor(negative, cv2.COLOR_BGR2RGB))
plt.title("Negative Image")
plt.axis("off")
plt.show()
```

Negative Image



```
roi = image[150:500, 150:500]

plt.figure(figsize=(7, 6))
plt.imshow(cv2.cvtColor(ro, cv2.COLOR_BGR2RGB))
plt.title("Region of Interest (ROI)")
plt.axis("off")
plt.show()
```

Region of Interest (ROI)

